

Ryan Charles Lingo, M.A.

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PROFESSIONAL EXPERIENCE

Software Development Specialist

January 2026 – Present
New Jersey

New Jersey Department of Environmental Protection

- Maintain and develop internal web applications supporting environmental regulatory workflows
- Build and update database-driven document generation systems with dynamic merge fields for automated letter production
- Work with stored procedures and database functions to support application logic and data integrity
- Optimize file compression workflows and provide technical support to resolve end-user application errors

GIS Database Engineer

January 2025 – September 2025
Remote

Excellent Technology Solutions (XTS) – United States Geological Survey

- Developed Python-based geoprocessing tools for transportation, graphics, and spatial datasets, automating workflows that supported national geospatial programs
- Managed and optimized geodatabases with ArcGIS Enterprise, Workflow Manager, and PostgreSQL/pgAdmin, leveraging traditional and branch versioning
- Built replica sync automation script with validation counts and automated email reporting; supported EC2 monitoring and sync integrity
- Contributed to Terraform-driven deployments of EC2 instances with mission-critical software
- Collaborated in Agile workflows using Jira and GitLab for task tracking, source control, and code reviews

GIS Developer/Automation Support

October 2020 – August 2023
Washington, DC

Leidos Inc. – Federal Aviation Administration

- Built a Python-based 3D airspace visualization system that mapped aircraft positions against satellite orbital paths using Skyfield and sun-angle geometry
- Enabled analysts to evaluate pilot-reported UAP sightings by testing line-of-sight intersections with Starlink constellation passes; work was reviewed by ODNI
- Co-authored abstract accepted for presentation at the 2023 ESRI User Conference (Abstract #1240)
- Supported Counter-UAS (C-UAS) initiatives and collaborated across multiple FAA lines of business
- Developed geoprocessing workflows and automated data analysis using Python and AWS tools

Geographer

February 2019 – October 2020
Washington, DC

United States Census Bureau

- Designed Python tools to support spatial analysis and data quality control for nationwide geographic programs
- Developed and implemented quality control systems to meet operational schedules and standards
- Completed advanced courses in ArcPy, Python, and Machine Learning to enhance geospatial analysis capabilities

Graduate Assistant

August 2016 – May 2018
Towson, MD

Department of Geography and Environmental Planning

- Assisted with GIS lab grading and research projects, including mapping land surface temperature for city planning
- Provided support for courses in GIS, Geography, and Climatology, contributing to academic development

Geospatial Research Consultant

May 2017 – September 2017
Greenbelt, MD

Science Systems and Applications Inc., NASA DEVELOP National Program

- Developed a global resiliency monitoring tool based on precipitation analysis for the Niger Water Resources project

Paleoclimatology Intern - Paleogeosciences Cataloguer

May 2014 – September 2014
Boulder, CO

NOAA - World Data Center for Paleoclimatology

- Collected and cataloged paleogeoscience resources for global database integration, authoring over 200 resource descriptions

EDUCATION

Master of Arts - Geography and Environmental Planning

Towson University - Towson, Maryland

Bachelor of Science - Geology

California University of Pennsylvania - California, Pennsylvania

Bachelor of Science - Environmental Earth Science

Minor: Meteorology

TECHNICAL SKILLS

GIS Software & Platforms

ArcGIS Enterprise, ArcGIS Pro, ArcGIS Desktop, ArcGIS Online, ArcGIS Workflow Manager, QGIS, TerrSet

Programming & Automation

Python, ArcPy, SQL, Terraform, JavaScript, HTML, CSS

Spatial Database Management

PostgreSQL/PostGIS, pgAdmin, Oracle SQL Developer, TOAD, Geodatabase Design, Branch Versioning

Cloud & Infrastructure

Amazon Web Services (AWS), EC2, Terraform, Cloud-based GIS Workflows

Data Analysis & Visualization

Tableau, Power BI, SPSS, Geoprocessing Workflows, Spatial Analysis, Data Quality Control

Python Libraries & Tools

ArcPy, PANDAS, Fiona, BOTO3, arcgis, urllib3, regex, json, git

Project Management & Collaboration

JIRA, GitLab, GitHub, Agile Workflows, Microsoft Office 365, SharePoint, Teams

Data Formats & Standards

Raster, Vector, KML, GeoJSON, Shapefile, XML, JSON-LD, Geospatial Data Standards

CERTIFICATIONS

AWS Certified Cloud Practitioner Foundational – December 2023

PUBLICATIONS & RESEARCH

Lingo, Ryan Charles. *Plausibly Deniable: A Documented Account of Government Retaliation*. Amazon Kindle, 2026.

Tomlin, J., El-Behaedi, R., Lingo, R., McCartney, S., Thieme, "Stress and Duress in the Sahel: Building Resiliency in Niger" Earthzine August (2017)

Anderson, David M., K. Horlick, and R. Lingo. "Share Your Geoscience Resources in the EarthCube Paleogeoscience Catalog." Past Global Changes, PAGES June (2014): 1, Web.

Conference Presentations: Long-term trend analysis and changes to the mean and variability of temperature in Pennsylvania from 1983 to 2014; American Association of Geographers, 2017

Conference Abstracts: Comparing Reports of Unidentified Aerial Phenomena to Starlink Satellite Orbits (Abstract #1240); ESRI User Conference, 2023

Research Projects: Baltimore City Urban Heat Island: A Problem Based Approach; Middle Atlantic Division of the American Association of Geographers, 2016